Chao-Ming Li

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0000-0002-9654-9123



Research Interests: Cosmic-ray transport, pulsar halos, and interstellar magnetic turbulence; numerical modeling of non-thermal emission and X-ray data analysis.

Education

09/2024 – present — **Deutsches Elektronen-Synchrotron DESY** — **Zeuthen, Germany**
Ph.D. candidate, Astrophysics — Supervisor: Andrew Taylor
Thesis: *Cosmic Ray Transport in the Giant Milky Way halo*
Expected Ph.D. completion: late 2027.

09/2021 – 06/2024 — **Nanjing University** — **Jiangsu, China**
M.Sc., Astronomy — Supervisor: Prof. Ruo-Yu Liu
Thesis: *Study of Very-High-Energy Gamma-ray Emission from Astrophysical Objects Related to Middle-Aged Pulsars*

09/2017 – 06/2021 — **Nanjing University** — **Jiangsu, China**
B.Sc., Astronomy

Research Publications

First-author Publications

1 **Li C.-M.**, Taylor A., 2026, ApJ, accepted. Galactic Cosmic Ray Transport in the Giant Circumgalactic Medium Halo
Contribution: Developed a numerical solver and Monte Carlo simulation code to investigate cosmic-ray transport in an extended circumgalactic halo and compare predictions with those of conventional small-halo models.

2 **Li C.-M.**, Liu R.-Y., Yan H., 2026, Phys. Rev. D, 113, L021303. Diagnosing interstellar magnetic turbulence with TeV pulsar halos
Contribution: Developed a theoretical interpretation of extended gamma-ray emission morphologies of pulsar halos in a coherent interstellar magnetic field and tested the predictions using numerical simulations.

3 **Li C.-M.**, Ge C., Liu R.-Y., 2023, ApJ, 949, 90. Revisiting the Chandra Observation on the Region of PSR J1809-193: Indication of the Existence of an X-Ray Halo and Implication for the Origin of HESS J1809-193.
Contribution: Led the reduction and analysis of Chandra observations and the numerical modeling of multiwavelength emission morphologies and spectra.

Co-authored Publications

4 Yan K., Liu R.-Y., Zhang R., **Li C.-M.**, Yuan Q., Wang X.-Y., 2024, NatAs, 8, 628. Insights from LHAASO and IceCube into the origin of the Galactic diffuse teraelectronvolt-petaelectronvolt emission

5 Wu Q.-Z., **Li C.-M.**, Liang X.-H., Ge C., Liu R.-Y., 2024, ApJ, 969, 9. Diagnosing the Particle Transport Mechanism in the Pulsar Halo via X-Ray Observations

6 Liang X.-H., **Li C.-M.**, Wu Q.-Z., Pan J.-S., Liu R.-Y., 2022, Univ, 8, 547. A PeVatron Candidate: Modeling the Boomerang Nebula in X-ray Band

7 Zhang Z.-J., Yan K., **Li C.-M.**, Zhang G. Q., Wang F. Y., 2021, ApJ, 906, 49. Intergalactic Medium Dispersion Measures of Fast Radio Bursts Estimated from IllustrisTNG Simulation and Their Cosmological Applications

Skills

Programming	Python, Fortran, C, Bash, LaTeX, R …
Data Reduction	Chandra, XMM-Newton, Swift, Fermi, LHAASO, …
Astronomy packages	HEASoft, XSPEC, PIMMS, DS9, TOPCAT …

Scientific Experience

Conference Presentations

2026
- γ **2026**, Heidelberg, Germany. Poster: Galactic Cosmic Ray Transport in the Giant Circumgalactic Medium Halo
- **UHE Symposium 2026**, Yuxi, China. Talk: Galactic Cosmic Ray Transport in the Giant Circumgalactic Medium Halo

2025
- **LHAASO Symposium 2025**. Hong Kong, China. Poster: Diagnosing interstellar magnetic turbulence with TeV pulsar halos

2023
- **ICRC 2023**, Nagoya, Japan. Talk: X-ray observation of HESS J1809-193: indication of an X-ray halo and implication for its γ-ray origin

Schools and Training

2025
- **ISAPP Summer School**, Lecce, Italy. High-energy cosmic rays as messengers: from space to the ground

2023
- **Research visit**, Yerevan, Armenia. Modeled the spectral energy distribution of the gamma-ray binary associated with LHAASO J2032+4127 in collaboration with Felix Aharonian and Dmitri Khangulyan.